Alex Crystal

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01 Summary

Generative AI engineer working across natural-language systems, topic modeling, knowledge graphs, forecasting, and misinformation analysis.

My work moves between research prototypes and applied engineering, with an emphasis on measurable behavior and tools people can actually use.

02 Experience

2025 — PRESENT
THE MITRE CORPORATION

Generative AI Engineer
Generative and Agentic AI Engineering
Built a hierarchical topic model and added topic nodes to an agentic knowledge graph for Kohai R6.

2020 — 2024
THE MITRE CORPORATION

Graduate Artificial Intelligence Intern
Human Language Technology

> **Summer Hackathon, 2023** · Placed first after augmenting training data and fine-tuning a transformer for phishing-site classification; the model achieved the competition's highest accuracy.

> **TextCraft, 2023** · Trained a Siamese network for authorship attribution, reaching 97% across 119 labels, and designed a reward signal for RL-fine-tuned language models.

> **S&TI 2.0, 2022** · Contributed to a science-and-technology knowledge graph and built deep topic models with TMNT to cluster academic papers.

> **ASIAS, 2022** · Fine-tuned multivariate aviation-forecasting models with Bayesian Hyperband optimization.

> **AM!TT, 2021** · Deployed a framework for characterizing adversarial disinformation tactics and techniques.

> **Bot Detector, 2020** · Implemented a RoBERTa discriminator and GPT-2 generator for social-media misinformation detection.

2019
BODESWELL INC.

Software Engineering Intern
Wrote Mocha unit tests that reached full coverage of the backend infrastructure.

03 Education

COMPLETED 2024

Northeastern University · Khoury College
MS, Artificial Intelligence
GPA 3.90

- Graduate Peer Pathways Mentor
- Khoury Research Apprentice: Generative AI to Empower Workers

COMPLETED 2022

University of Rochester · Goergen Institute
BS, Data Science
High Distinction · Cum Laude

- Wilder Leadership Scholar
- Teaching assistant for Data Science Capstone + Practicum and Tools for Data Science (2×)
- Transformed 300GB of NASA data into microeconomic trend forecasts with MacroXStudio Inc.

04 Research & projects

2023

CherÜb
NORTHEASTERN UNIVERSITY
Built an LLM application with Zapier, Pinecone, OpenAI, and Slack.

2023

Bach-in-Business ↗
NORTHEASTERN UNIVERSITY
Explored AI music generation with biaxial LSTMs.

2022

Research Assistant
UNIVERSITY OF ROCHESTER · DEPARTMENT OF MATHEMATICS
Conducted deep-learning research in time-series forecasting through the TRIPODS REU.

2020 — 2021

Research Assistant
UNIVERSITY OF ROCHESTER MEDICAL CENTER
Researched biological-network analysis methods and contributed to the mBONITA package ↗.

05 Publication

JOURNAL OF PROTEOME RESEARCH · 2023
Executable Network Models of Integrated Multiomics Data
Palshikar MG, Min X, Crystal A, Meng J, Hilchey SP, Zand MS, Thakar J. Volume 22, Issue 5, 1546–1556.
PubMed ↗ DOI ↗

06 Skills

Languages	Python · Java · SQL · HTML · CSS · JavaScript · R
Machine learning	Transformers · PyTorch · TensorFlow / Keras · Ray / RLlib · Gym · NLTK · OpenCV · MXNet
Data & infrastructure	AWS · Slurm · Hugging Face · MongoDB · MySQL · Git
Application tools	Flask · Django · Matplotlib · Zapier · Jira · Confluence · IntelliJ IDEA

07 Leadership & highlights

ATHLETICS
Club Running National Champion

SOCCER
League Champion · Vice President

LEADERSHIP
Chabad Rochester · President

LEADERSHIP
Delta Upsilon · Vice President

ENDURANCE
IRONMAN 70.3

SERVICE
STEM Initiative & New Balance Nationals volunteer